SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1106

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 6 **Total Marks:30**

Annexure A

Scenario-Based

Code of Conduct:

I D Raghul 192511115 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-Based

1. Use Case Diagram Scenario

A **Smart Cinema Ticket Booking System** allows customers to search movies, select seats, book tickets, cancel bookings, and make online payments. Theater managers update movie schedules and seat availability, while administrators manage users and system settings. The interactions between actors and system functions are not clearly defined.

Based on this scenario, analyze the system requirements, identify all actors and use cases, and design a **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to ensure complete system functionality.

(5 Marks)

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2. Class Diagram Scenario

A **Smart College Management System** consists of entities such as Student, Faculty, Department, Course, Attendance, and Examination. The existing design does not clearly define class attributes, operations, or relationships among these entities.

Based on this scenario, identify the required classes, define suitable attributes and methods, establish relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a detailed **Class Diagram**.

(5 Marks)

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3. Sequence Diagram Scenario

In an **Online Railway Reservation System**, a passenger searches for trains, selects a seat, books a ticket, makes payment, and receives an e-ticket. The communication among the passenger, reservation system, payment gateway, and database is incomplete.

Based on this scenario, analyze the sequence of operations, identify participating objects and message flows, and design a **Sequence Diagram** representing the complete interaction process.

(5 Marks)

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4. Activity Diagram Scenario

A **Smart Insurance Claim Processing System** allows customers to submit insurance claims, upload supporting documents, verify claim details, approve or reject claims, and receive compensation. The workflow does not clearly represent decision-making or concurrent activities.

Based on this scenario, analyze the workflow, identify decision points and parallel activities, and design an optimized **Activity Diagram** using decision nodes, forks, and joins.

(5 Marks)

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5. State Diagram Scenario

A **Smart Washing Machine System** operates through states such as Idle, Water Filling, Washing, Rinsing, Spinning, Paused, and Completed. The transitions between these operational states are not clearly specified.

Based on this scenario, identify all possible states, triggering events, and transitions, and construct a **State Transition Diagram** that accurately represents the washing machine's behavior.

(5 Marks)

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6. Deployment Diagram Scenario

A **Smart Energy Management System** uses smart electricity meters, edge gateways, cloud servers, utility monitoring dashboards, and mobile applications to monitor energy consumption and control power distribution. The deployment architecture does not clearly illustrate hardware nodes, deployed software artifacts, or communication links.

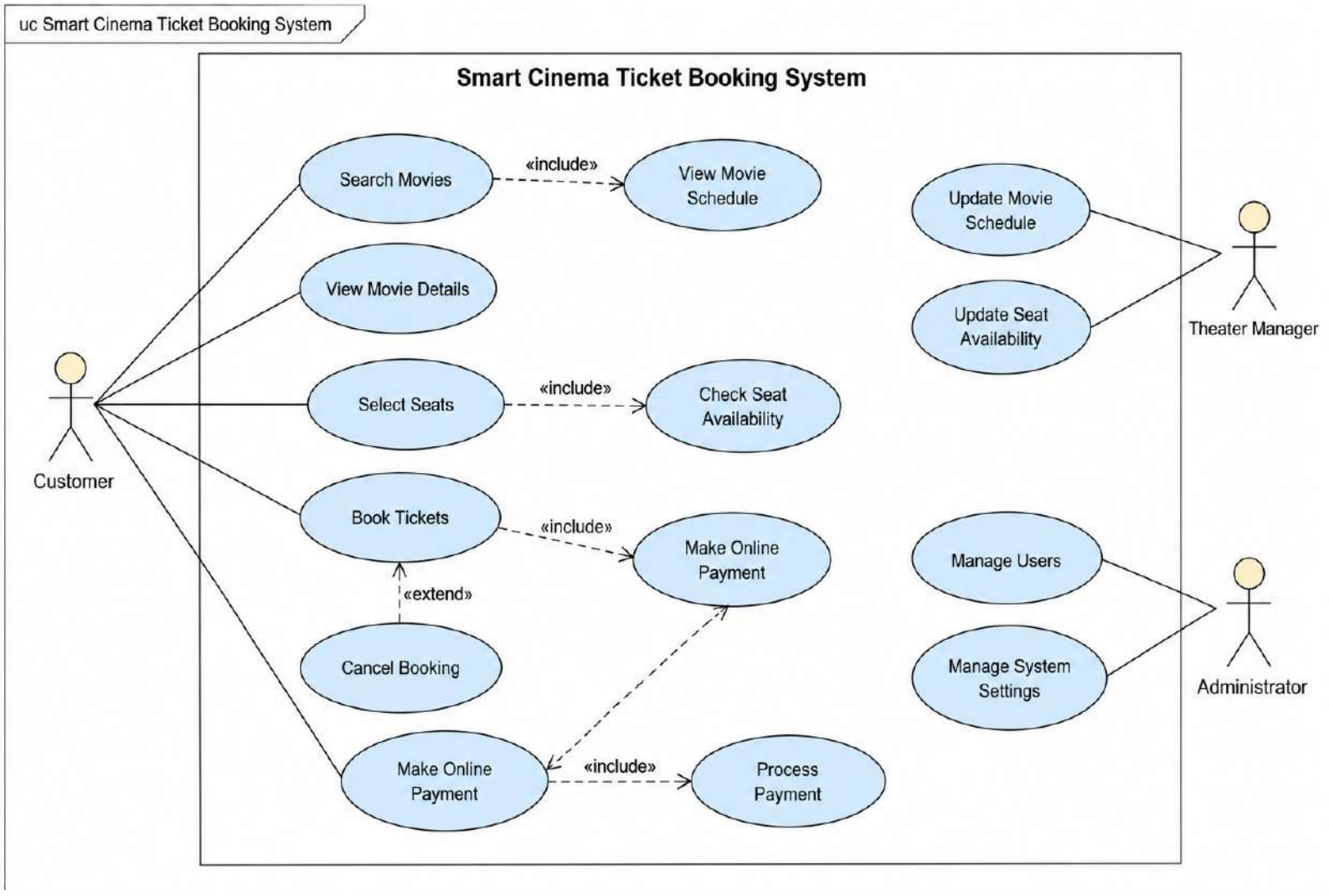
Based on this scenario, analyze the deployment environment, identify all hardware nodes and software artifacts, and design a **Deployment Diagram** showing the physical architecture and communication links.

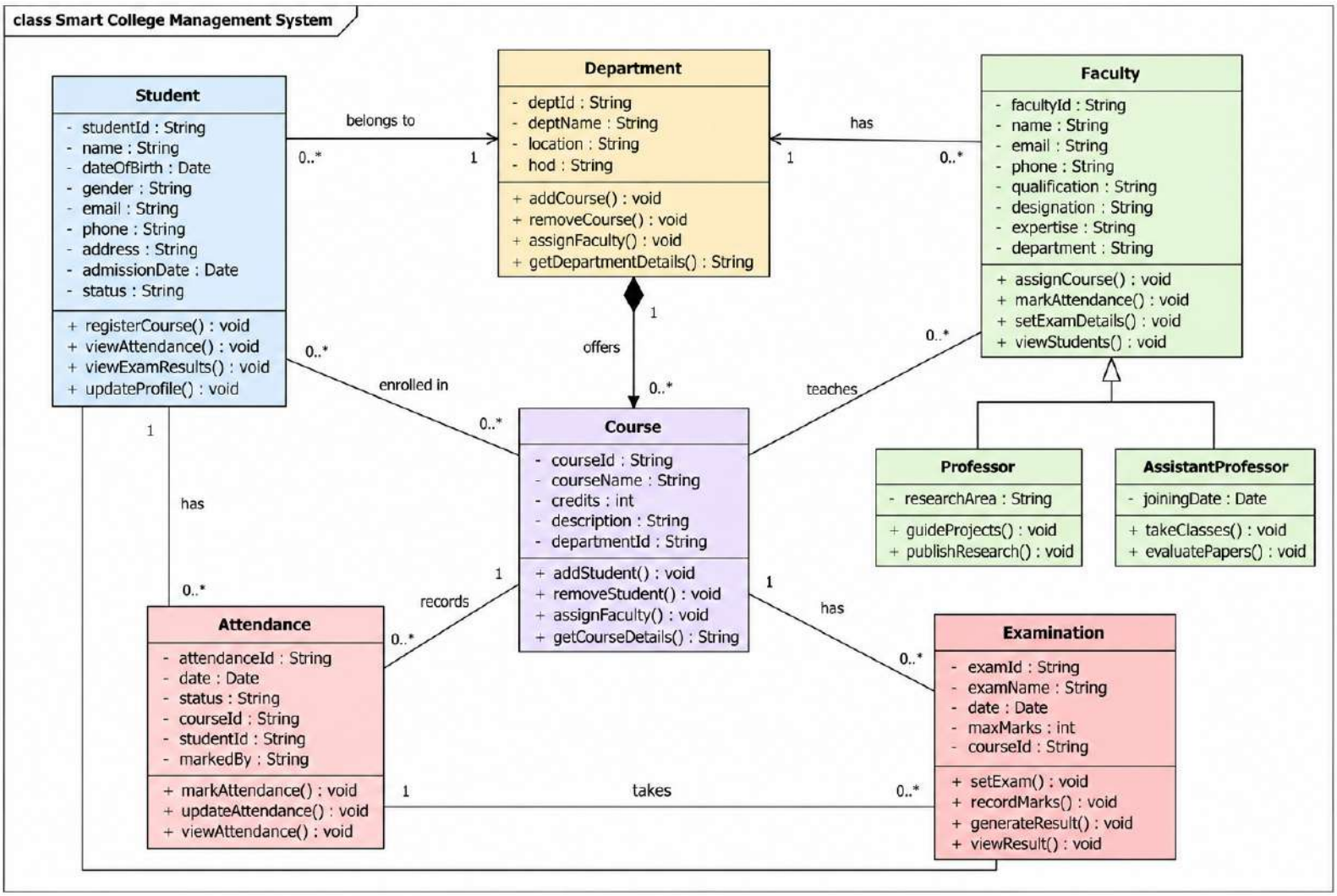
(5 Marks)

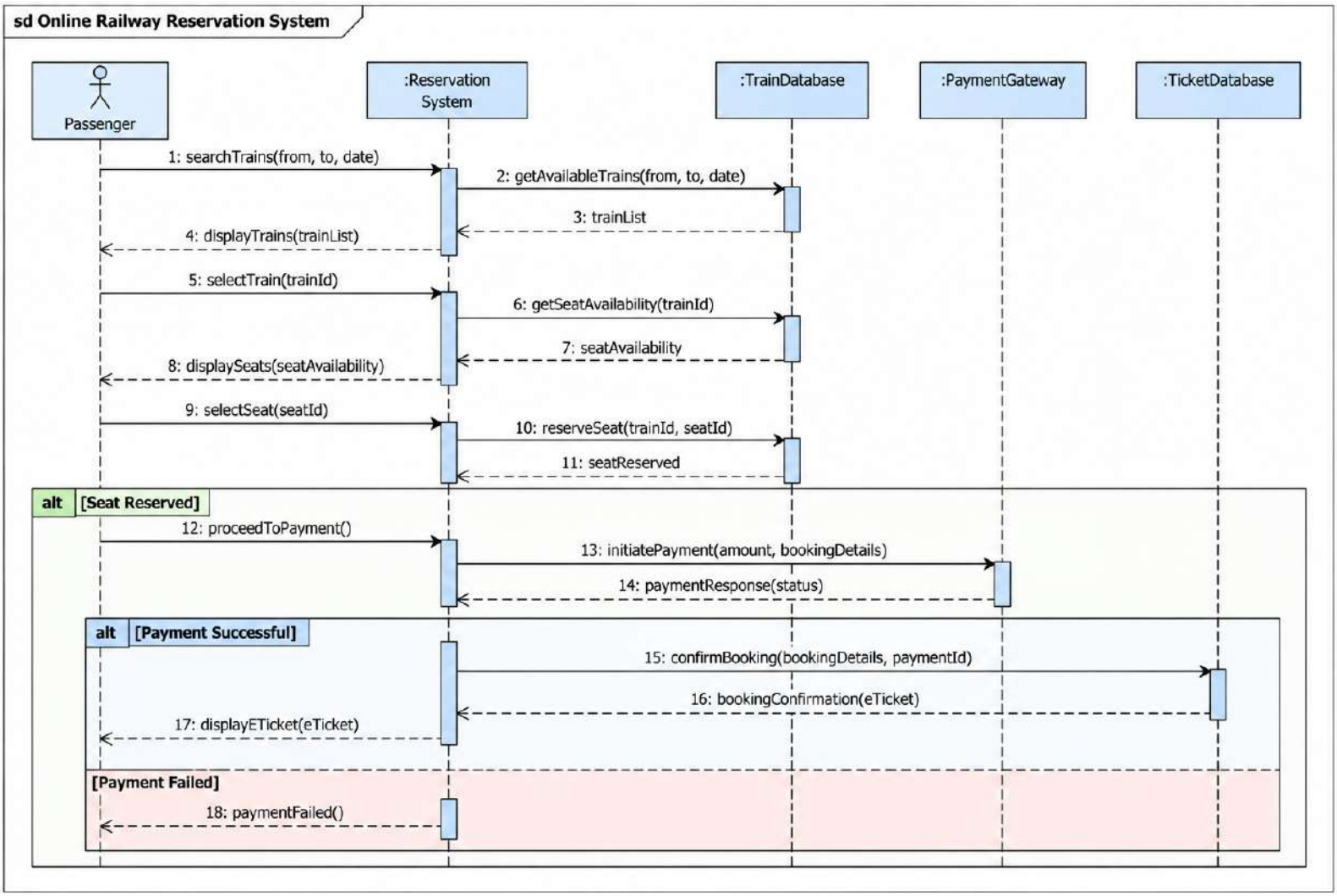
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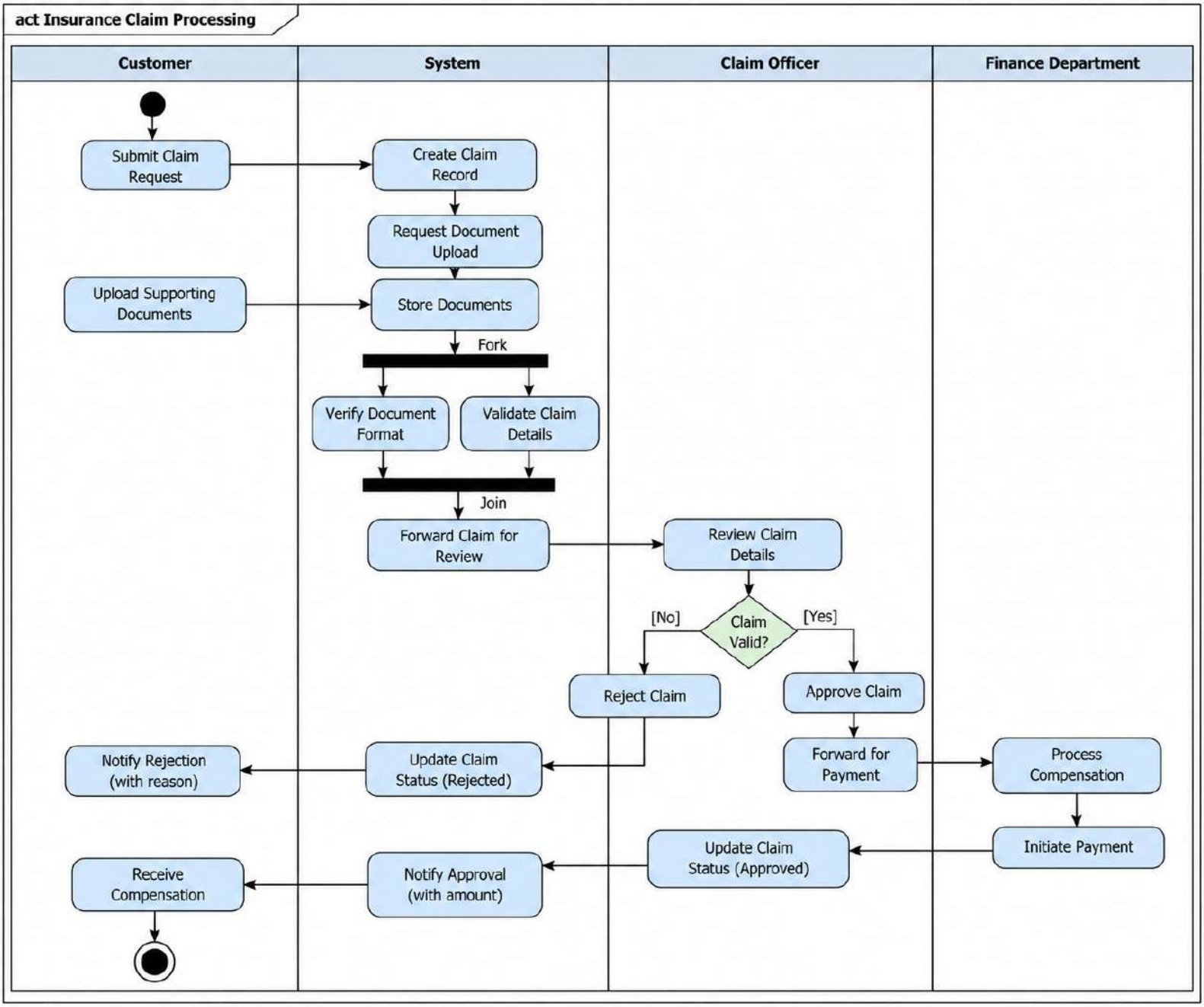
Scenario-Based

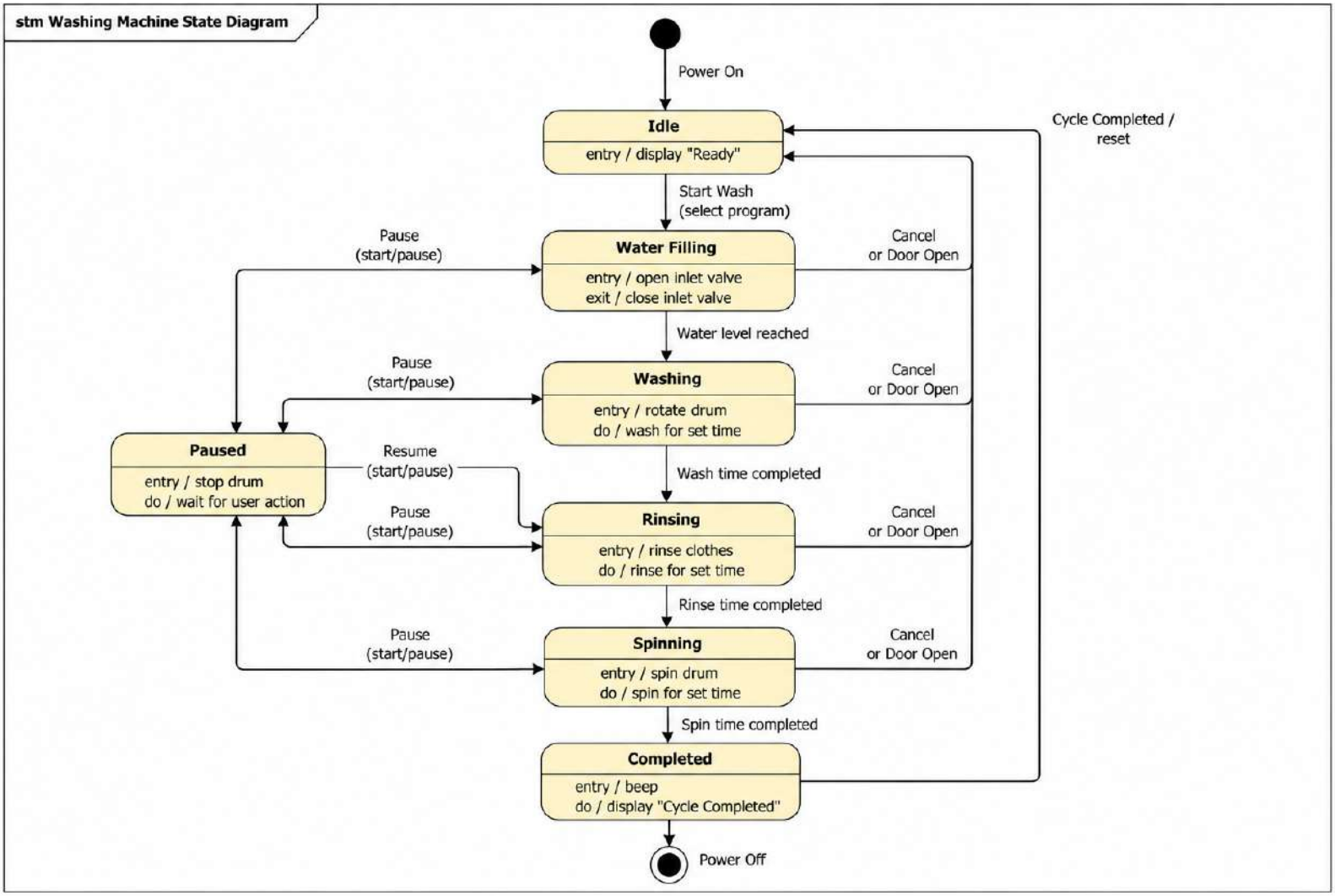
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

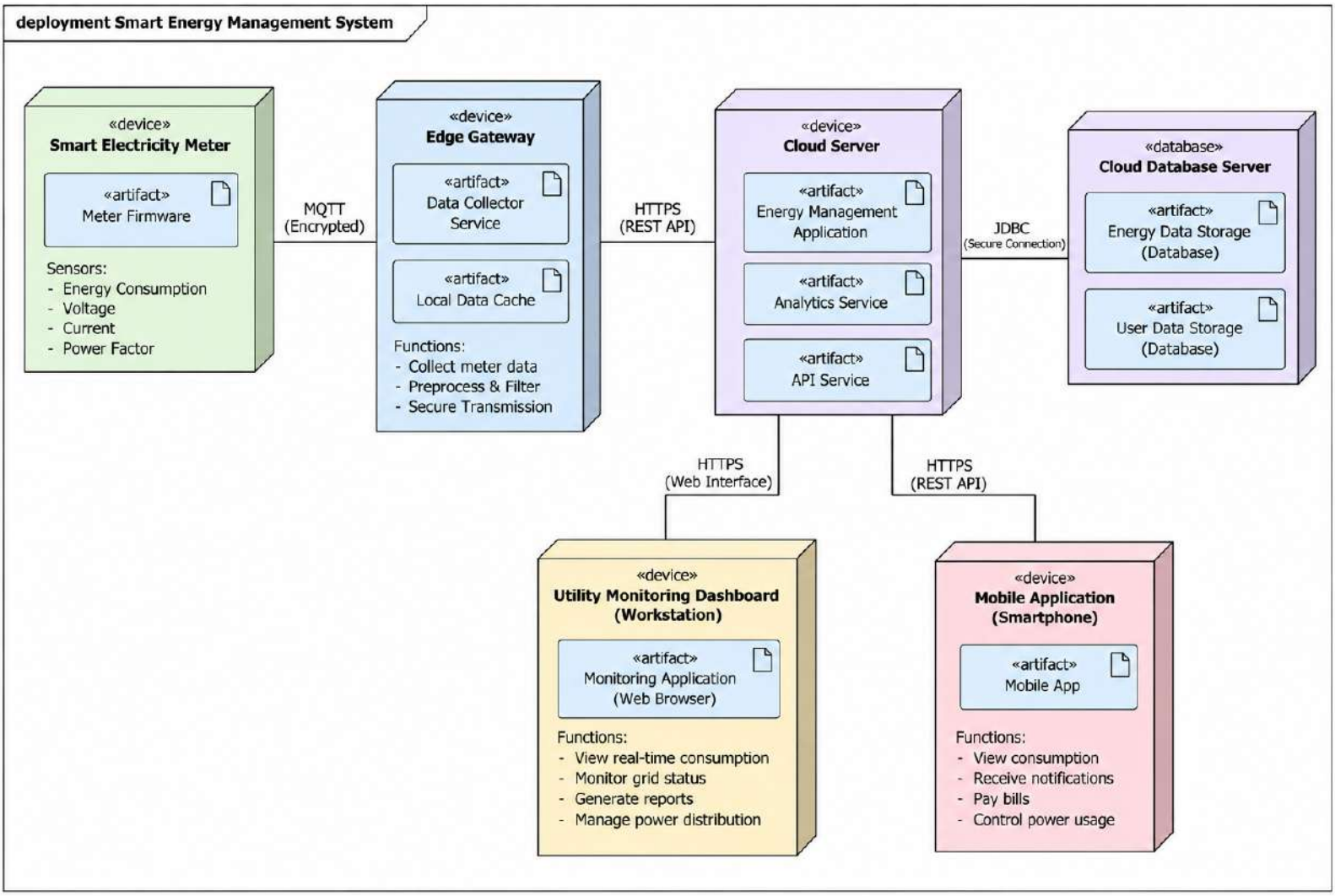

uc Smart Cinema Ticket Booking System
Smart Cinema Ticket Booking System
Search Movies
«include»
View Movie Schedule
View Movie Details
Select Seats
«include»
Check Seat Availability
Book Tickets
«include»
Make Online Payment
«extend»
Cancel Booking
Make Online Payment
«include»
Process Payment
Update Movie Schedule
Update Seat Availability
Theater Manager
Manage Users
Manage System Settings
Administrator
Customer


class Smart College Management System
Student
- studentId : String
- name : String
- dateOfBirth : Date
- gender : String
- email : String
- phone : String
- address : String
- admissionDate : Date
- status : String
+ registerCourse() : void
+ viewAttendance() : void
+ viewExamResults() : void
+ updateProfile() : void
Department
- deptId : String
- deptName : String
- location : String
- hod : String
+ addCourse() : void
+ removeCourse() : void
+ assignFaculty() : void
+ getDepartmentDetails() : String
Faculty
- facultyId : String
- name : String
- email : String
- phone : String
- qualification : String
- designation : String
- expertise : String
- department : String
+ assignCourse() : void
+ markAttendance() : void
+ setExamDetails() : void
+ viewStudents() : void
Course
- courseId : String
- courseName : String
- credits : int
- description : String
- departmentId : String
+ addStudent() : void
+ removeStudent() : void
+ assignFaculty() : void
+ getCourseDetails() : String
Professor
- researchArea : String
+ guideProjects() : void
+ publishResearch() : void
AssistantProfessor
- joiningDate : Date
+ takeClasses() : void
+ evaluatePapers() : void
Attendance
- attendanceId : String
- date : Date
- status : String
- courseId : String
- studentId : String
- markedBy : String
+ markAttendance() : void
+ updateAttendance() : void
+ viewAttendance() : void
Examination
- examId : String
- examName : String
- date : Date
- maxMarks : int
- courseId : String
+ setExam() : void
+ recordMarks() : void
+ generateResult() : void
+ viewResult() : void
belongs to
0..*
1
has
1
0..*
offers
1
0..*
enrolled in
0..*
0..*
teaches
0..*
1
has
0..*
records
1
0..*
1
has
0..*
1
takes
0..*


sd Online Railway Reservation System
Passenger
:Reservation System
:TrainDatabase
:PaymentGateway
:TicketDatabase
1: searchTrains(from, to, date)
2: getAvailableTrains(from, to, date)
3: trainList
4: displayTrains(trainList)
5: selectTrain(trainId)
6: getSeatAvailability(trainId)
7: seatAvailability
8: displaySeats(seatAvailability)
9: selectSeat(seatId)
10: reserveSeat(trainId, seatId)
11: seatReserved
alt
[Seat Reserved]
12: proceedToPayment()
13: initiatePayment(amount, bookingDetails)
14: paymentResponse(status)
alt
[Payment Successful]
15: confirmBooking(bookingDetails, paymentId)
16: bookingConfirmation(eTicket)
17: displayETicket(eTicket)
[Payment Failed]
18: paymentFailed()


act Insurance Claim Processing
Customer
System
Claim Officer
Finance Department
Submit Claim Request
Create Claim Record
Request Document Upload
Upload Supporting Documents
Store Documents
Fork
Verify Document Format
Validate Claim Details
Join
Forward Claim for Review
Review Claim Details
[No]
Claim Valid?
[Yes]
Reject Claim
Approve Claim
Notify Rejection (with reason)
Update Claim Status (Rejected)
Forward for Payment
Process Compensation
Initiate Payment
Update Claim Status (Approved)
Receive Compensation
Notify Approval (with amount)


stm Washing Machine State Diagram
Power On
Idle
entry / display "Ready"
Start Wash
(select program)
Water Filling
entry / open inlet valve
exit / close inlet valve
Water level reached
Washing
entry / rotate drum
do / wash for set time
Wash time completed
Rinsing
entry / rinse clothes
do / rinse for set time
Rinse time completed
Spinning
entry / spin drum
do / spin for set time
Spin time completed
Completed
entry / beep
do / display "Cycle Completed"
Power Off
Paused
entry / stop drum
do / wait for user action
Pause
(start/pause)
Resume
(start/pause)
Cancel
or Door Open
Cycle Completed /
reset


deployment Smart Energy Management System
«device»
Smart Electricity Meter
«artifact»
Meter Firmware
Sensors:
- Energy Consumption
- Voltage
- Current
- Power Factor
MQTT
(Encrypted)
«device»
Edge Gateway
«artifact»
Data Collector
Service
«artifact»
Local Data Cache
Functions:
- Collect meter data
- Preprocess & Filter
- Secure Transmission
HTTPS
(REST API)
«device»
Cloud Server
«artifact»
Energy Management
Application
«artifact»
Analytics Service
«artifact»
API Service
JDBC
(Secure Connection)
«database»
Cloud Database Server
«artifact»
Energy Data Storage
(Database)
«artifact»
User Data Storage
(Database)
HTTPS
(Web Interface)
HTTPS
(REST API)
«device»
Utility Monitoring Dashboard
(Workstation)
«artifact»
Monitoring Application
(Web Browser)
Functions:
- View real-time consumption
- Monitor grid status
- Generate reports
- Manage power distribution
«device»
Mobile Application
(Smartphone)
«artifact»
Mobile App
Functions:
- View consumption
- Receive notifications
- Pay bills
- Control power usage